

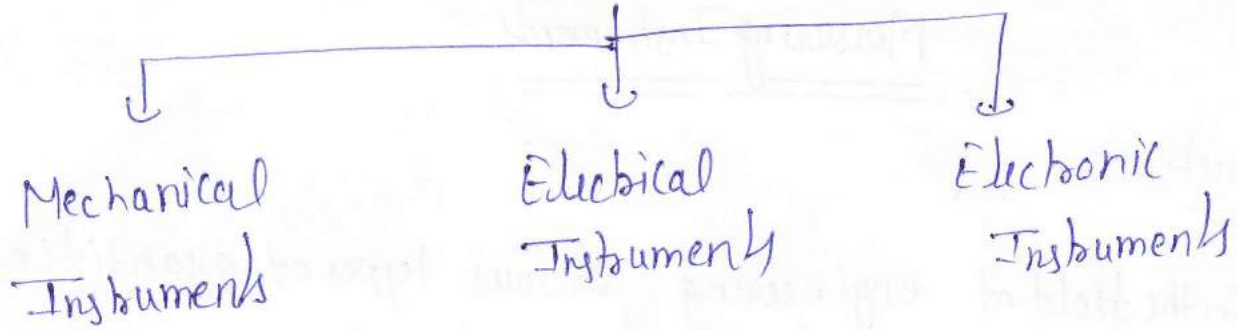
Measuring InstrumentsIntroduction:

- In the field of engineering, various types of quantities such as physical, chemical, Mechanical, thermal & electrical are involved.
- these quantities are required to be measured by different methods using different types of instruments.
- Electrical Measuring Instruments are widely used in the modern engineering world because of their accuracy, convenience & reliability.
- These instruments are not only used to measure the electrical quantities, but are also used to measure other non-electrical quantities such as temperature, strain, deformation, velocity & pressure with the help of transducers.

Instruments & their Classification:-

- A device used to measure some quantity is known as Instrument
- An instrument is used to determine the magnitude of given quantity.

Classification



→ Here we are interested in electrical measuring instruments so we will discuss about them in detail.

Electrical Instruments

→ In electrical system, there are various electrical quantities, such as current, voltage, power, energy, frequency, power factor, resistance, inductance & capacitance that are required to be measured.

→ To measure these quantities, various instruments such as ammeter, voltmeter, wattmeter, energymeter, frequency meter, power factor meter & ohm-meter are used. These instruments are known as electrical instruments.

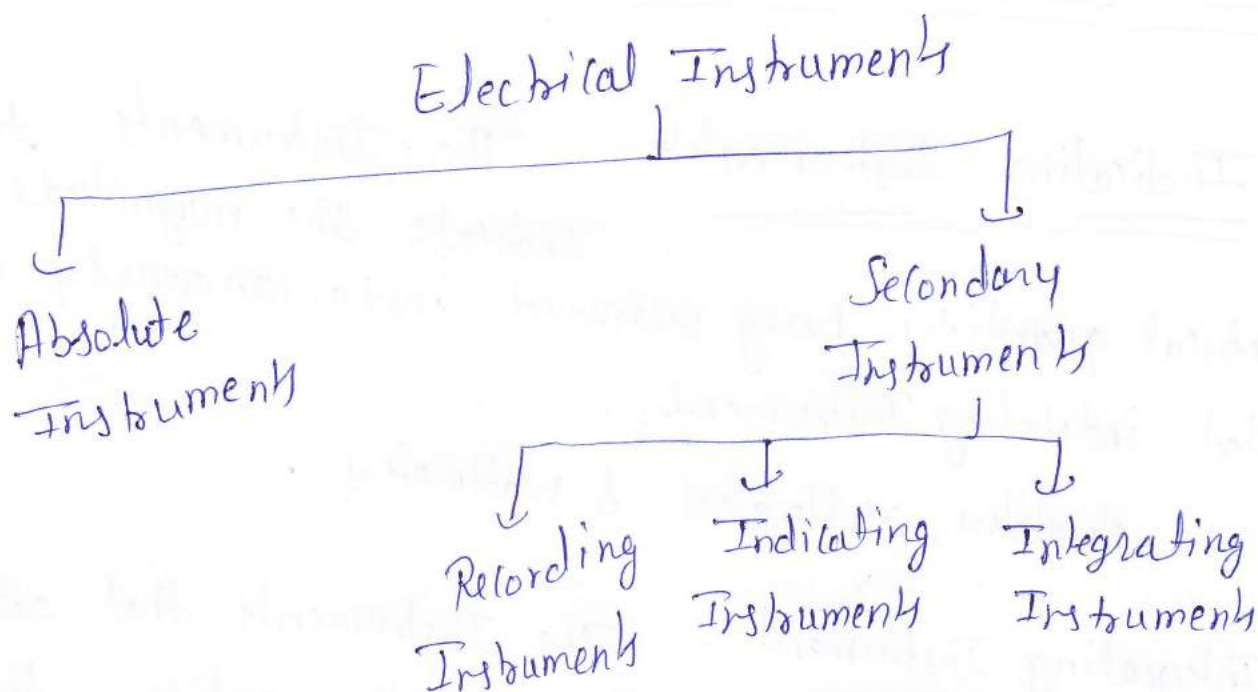
→ The instruments, such as ammeter, voltmeter, energymeter and megger, are used to measure electrical quantities and are called electrical instruments.

Classification of electrical Instruments:-

③

① ~~Ab~~

Classification:



① Absolute Instruments:-

→ The Instruments that gives the value of the quantity to be measured in term of constant of the Instruments are called absolute Instruments.

→ such Instruments do not Require any previous Calibration

Example → tangent galvanometer.

② Secondary Instruments:-

The Instruments that determine the electrical quantity to be measured directly in terms of deflection are called secondary Instruments.

→ Such Instruments are calibrated with standard Instruments that have already been calibrated before

using them. These Instruments ~~the~~ are Generally (4)
used in practical life.

Types of Secondary Instruments:-

1) Indicating Instruments:- The Instruments that indicate the magnitude of electrical quantity being measured instantaneously are called indicating Instruments.

eg:-> Ammeters, Voltmeters & Wattmeters.

2) Integrating Instruments:- The Instruments that add up the electrical quantity, that is, electrical energy & measure the total energy (in kWh) in a given period are called integrating instruments.

→ eg:-> Energy meters.

3) Recording Instruments:- The Instruments that give a continuous record of the variation of the electrical quantity being measured are called Recording Instruments.

eg:-> Electrocardiograph (ECG), Graph based on a paper etc.

Effects utilised in electrical instruments to produce deflecting torque:-

→ The principle of operation of electrical instruments depends upon various effects produced by electric current & voltage.

Effects utilized in various Electrical Instruments:-

Effect produced

- 1) Magnetic effect
- 2) Electromagnetic ^{Induction} effect

3) Electrodynamic effect

4) Electrostatic effect

5) Heating or thermal effect

6) Chemical effect

Instruments in which this effect is generally utilised

Ammeters & voltmeters.

Ammeters, voltmeters, & wattmeter & energy meter, usually wattmeter & energy meters.

Ammeters, voltmeters & usually wattmeter usually wattmeter.

Voltmeters only

Ammeters & voltmeters

DC ampere-hour meters

⊗ Essentials of Indicating Instruments:-

In Indicating Instruments, a pointer moves over the calibrated scale to indicate the magnitude of electrical quantity being measured. The pointer is attached

to a moving system pivoted in jewelled bearings. ⑥
→ The forces or torques Required for satisfactor operation of indicating Instruments.

1) Deflecting torque:-

→ The deflecting or operating torque (T_d) is produced by making use of any one of the effects such as magnetic, electrodynamics & electromagnetic induction.

→ The deflecting torque is Required to move the moving system from zero position when the Instrument is connected in the circuit to measure the electrical quantity.

2) Controlling torque:-

The controlling or Restoring torque (T_c) oppose the deflecting torque and increase with the deflection of the moving system.

→ The pointer is brought to Rest at a position where the two opposing torques (i.e. deflecting torque & controlling torque) are equal.

→ If this torque was not provided, the pointer would continue to move indefinitely and the deflection shall be independent of the value of electrical quantity being measured.

→ The other function of the controlling torque is ⑦ to bring the pointer back to zero, when the instrument is removed from the circuit or when the deflecting torque acting in the instrument is removed.

→ If this torque was not provided, the pointer once deflected ~~was~~ would not return to zero position on removing the deflecting torque.

→ This torque is either obtained by a spring or by gravity in the indicating instruments.

→ In Spring Control, one or two phosphor bronze spiral ~~bars~~ -springs are attached to the moving spindle. The other ends of the springs are attached to the fixed body or frame. In this controlling torque (T_c) is directly proportional to the angle of deflection (θ), that is $T_c \propto \theta$

→ In Gravity Control: A small adjustable weight is attached to the moving system in such a way that due to gravity, it ~~rise~~ tries to bring the pointer to zero position when it is deflected. In this case the controlling torque (T_c) is proportional to the sine of angle of deflection (θ), i.e. $T_c \propto \sin \theta$.

Damping torque:-

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- When deflecting torque is applied to the moving system, it deflects the pointer.
- While the controlling torque ~~controls~~ controls the deflection & tries to stop the pointer at its final position.
- However due to inertia, the pointer oscillates around its final ~~opposite~~ position before coming to rest.
- These oscillations are undesirable because it causes delay in taking the reading. Then what to do?
- In order to avoid these oscillations & to bring the pointer quickly to its final position, damping torque is provided.
- The damping torque opposes the movement (forward or backward) of the pointer & operates only when the system is moving.
- In fact, damping torque acts like a brake on the moving system.

Methods of providing damping torque:-

- 1) Air friction damping
- 2) fluid friction damping
- 3) Eddy current damping

Classification of Instruments

⑨

Moving Iron

PMMC Moving Coil

Dynamometer
type

Induction
type
Instruments

① PMMC:-

PMMC $\rightarrow$ permanent Magnet Moving Coil Instruments

Introduction:-

- $\rightarrow$ These instruments are very sensitive & accurate. These can be used only on DC as Voltmeter & ammeter.
- $\rightarrow$ The scale of such Instrument is uniform.

Principle:-

The basic principle of PMMC Instrument is that, when a current-carrying conductor is placed in a magnetic field, a mechanical force is exerted on the conductor.

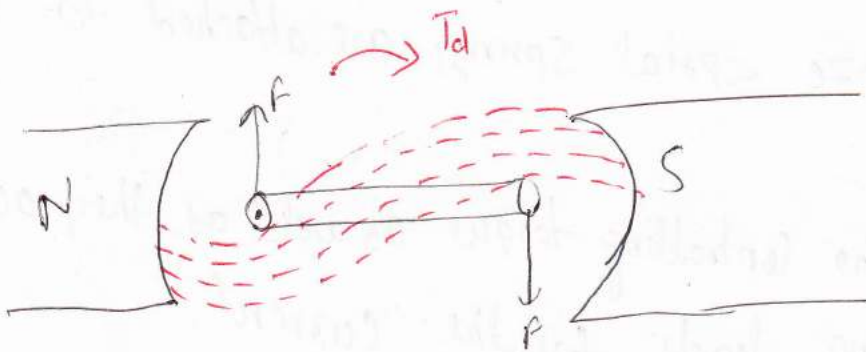
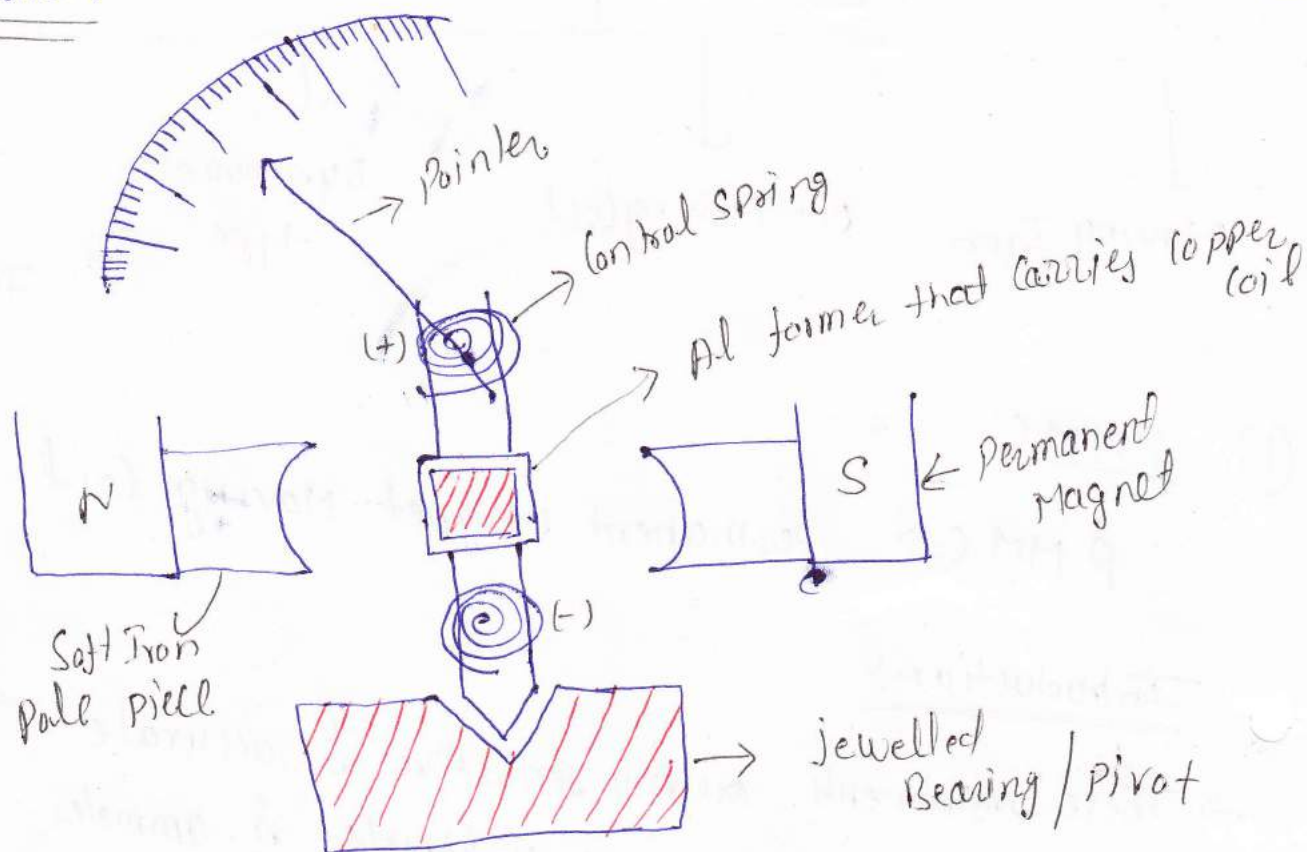


fig $\rightarrow$ force exerted on a current carrying conductor placed in a magnetic field.

Construction:-

(10)



- It consists of permanent Magnet, a light Rectangular coil of many turns of fine wire wound on a light Al former & a cylindrical iron core (a stationary part) inserted in between the coils sides.
- The coil is mounted on the spindle and acts as the moving element.
- Two phosphor bronze spiral springs are attached to the spindle.
- Springs provide the controlling torque as well as they act as incoming & outgoing leads for the current.
- Eddy current damping is provided by the Al former over which the operating coil is wound.

Working:-

(11)

- When the Instrument is connected in the circuit, a current flows through the operating coil mounted on the spindle.
- Since the coil is placed in the strong Magnetic field of Permanent Magnets, a force is exerted on the current-carrying conductors of the coil that produces deflecting torque.
- Thus, the pointer attached to the spindle deflects over the calibrated scale.
- If ~~the~~ current in the coil is Reversed, the direction of deflecting torque will also be Reversed because field produced by the permanent Magnets does not change. This will give an opposite direction of Rotation. Thus the instrument can-not be used on AC & it can only be used on DC.

Deflecting Torque:-

- The deflecting torque T_d depends upon the force acting on the coil sides. Let
- B = flux density in Tesla in the air gap
- l = effective length of each coil side in metre.
- N = number of turns of the coil
- x = distance in meter between centre of the coil & forces

I = current flowing through the coil in ampere.
force acting on each coil side

$$F = BILN$$

Deflecting torque,

$$T_d = 2fr = 2BILNr \text{ (Nm-unit)}$$

$\therefore$ Since all other quantities are constant, except I

$$T_d \propto I$$

The Controlling Torque is provided by the springs

$$T_c \propto \theta \quad (\text{where } \theta \text{ is angle of deflection})$$

At steady position of deflection

$$T_c = T_d$$

$$\theta \propto I$$

Since deflection θ is proportional to the operating current flowing through the coil & therefore the scale of such instruments is uniform.

Advantages:-

- ① These instruments have uniform scale.
- ② Very effective; the reliable eddy current damping is provided
- ③ No hysteresis loss, as the former is of Al.
- ④ Low power consumption
- ⑤ Very accurate & reliable

Disadvantages:-

- ① These instruments can-not be used for AC Measurements.
- ② These are costlier in comparison to Moving iron instruments
- ③ Some errors are also caused due to ageing of control springs and permanent magnets.

Errors:-

- ① Weakening of stiffness of springs due to ageing & rise in temperature.
- ② Weakening of field produced by permanent magnet due to ageing & temperature effects.

Range:-

DC - Ammeters:-

without shunt $\rightarrow$ 0-5 μ A to 0-50 μ A
 with ~~out~~ internal shunts up $\rightarrow$ 0-200 A
 " external " " $\rightarrow$ 0-5000 A

DC Voltmeter:-

- ① With out series Resistance or multiplier, 0-50mV
- ② with series Resistance, up to, 0-3000V

Application:-

① Ammeter

② Voltmeter

⊗ Extension of Range of Ammeters & Voltmeters:-

→ The Moving coil instruments can carry maximum currents of about 50mA safely & the potential drop across the moving coil is about 50mV.

→ However, in practice, heavy currents & voltages are Required to be measured.

→ There are four common devices used for extending the range of the instruments, namely shunts, multipliers, current transformer & potential transformers.

→ The extension of range of current and potential transformer is employed for measurement of very high AC currents and voltages in power system.

① Shunts:- These are used for the extension of range of ⁽¹⁵⁾ ammeter. Shunt is a Resistance of small value Just like a strip, having minimum temperature co-efficient.
→ It is always connected in parallel with the ammeter whose range is to be extended. The combination is connected in series with the circuit whose current is to be measured.

② Multipliers:- These are used for the extension of range of voltmeters. Multiplier is a non-inductive high Resistance usually carbon Resistors connected in series with the instrument whose range is to be extended.
→ The combination is connected across the circuit whose voltage is to be measured.

⊗ Extension of Ammeter Range:-

The current range of a DC moving coil ammeter is extended by connecting a shunt R_s (low Resistance) across the coil.

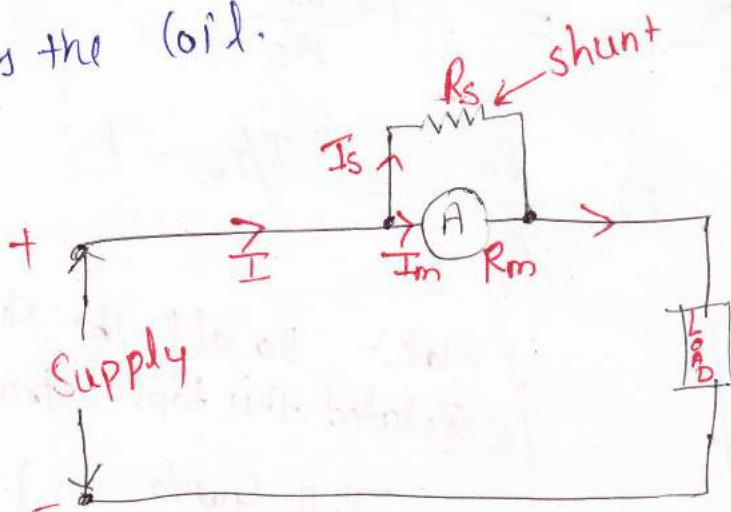


fig → Circuit for extension of ammeter Range.

Let

 I = Current to be measured I_m = full scale deflection current of ammeter I_s = shunt current R_m = Resistance of ammeter R_s = Shunt Resistance

from figure, we have

$$I = I_m + I_s \rightarrow \text{eqn (1)}$$

$$I_s = I - I_m \rightarrow \text{eqn (2)}$$

And ~~$I_s = I$~~ Voltage drop across R_s & R_m is equal
bcz both are connected in parallel.

$$I_s R_s = I_m R_m \rightarrow \text{eqn (3)}$$

$$(I - I_m) R_s = I_m R_m$$

$$\left(\because I_s = I - I_m \right) \\ (\text{see eqn (2)})$$

$$I R_s - I_m R_s = I_m R_m$$

$$I R_s = I_m (R_m + R_s)$$

$$\frac{I}{I_m} = \frac{R_m + R_s}{R_s} = 1 + \frac{R_m}{R_s}$$

$$\frac{I}{I_m} = 1 + \frac{R_m}{R_s} \Rightarrow \frac{R_m}{R_s} = \frac{I}{I_m} - 1$$

$$\boxed{R_s = \frac{R_m}{\frac{I}{I_m} - 1}}$$

Note:- Do all the Numericals
Related this topic from

J. B Gupta

Numericals:- (Ammeter extension Range)

(17)

Q:-1 A moving coil ammeter has a full-scale deflection of $50 \mu A$ and a coil Resistance of 1000Ω . What will be the value of the shunt Resistance Required for the instrument to be converted to Read a full-scale Reading of $1 A$?

Solⁿ

Full-scale deflection current, $I_m = 50 \times 10^{-6} A$

Instrument Resistance, $R_m = 1000 \Omega$

total current to be measured, $I = 1 A$

Value of shunt Resistance Required, $R_s = \frac{R_m}{I/I_m - 1}$

$$R_s = \frac{1000}{\frac{1}{50 \times 10^{-6}} - 1} = 0.0500025 \Omega$$

$$\boxed{R_s = 0.0500025 \Omega} \quad \underline{\text{Ans}}$$

Q:-2 The full-scale deflection current of an ammeter is $1 mA$ and its internal Resistance is 100Ω . If this meter is to have full-scale deflection at $5 A$, what is the value of the shunt Resistance to be used?

Solⁿ

Full-scale deflection current, $I_m = 1 mA = 0.001 A$

Instrument Resistance, $R_m = 100 \Omega$

total current to be measured, $I = 5 A$

Shunt Current, $I_s = I - I_m$

(18)

$$I_s = 5 - 0.001 = 4.999 A$$

$$\text{Value of Shunt Resistance, } R_s = \frac{I_m R_m}{I_s} = \frac{0.001 \times 100}{4.999} \quad (\text{see eqn (3)})$$

$$\boxed{R_s = 0.020004 \Omega} \quad \underline{\text{Ans}}$$

Q.13 Design a multi-Range ammeter with ranges of 1A, 5A, 25A & 125A employing individual shunts in each case. A personal movement with coil internal Resistance of 730Ω & a full scale current of 5 mA is available.

Solⁿ

$$\text{Meter Resistance, } R_m = 730 \Omega$$

$$\text{Meter Current, } I_m = 5 \text{ mA} = 0.005 A$$

$$\text{Shunt Current, } I_s = I - I_m$$

$$\text{Shunt Resistance, } R_s = \frac{I_m R_m}{I_s} = \frac{I_m R_m}{(I - I_m)} \quad (\text{using eqn (3)})$$

$$\text{for } 1A \text{ range, } R_{s1} = \frac{5 \times 10^{-3} \times 730}{(1 - 0.005)} = 3.668 \Omega$$

$$\text{for } 5A \text{ range, } R_{s2} = \frac{5 \times 10^{-3} \times 730}{(5 - 0.005)} = 0.7307 \Omega$$

$$\text{for } 25A \text{ range, } R_{s3} = \frac{5 \times 10^{-3} \times 730}{(25 - 0.005)} = 0.146 \Omega$$

$$\text{for } 125A \text{ range, } R_{s4} = \frac{5 \times 10^{-3} \times 730}{(125 - 0.005)} = 0.0292 \Omega$$

⊗ Extension of Voltmeter Range:-

(19) ~~10~~

→ The voltage range of a DC moving coil voltmeter is extended by connecting a multiplier R (high Resistance) in series with coil.

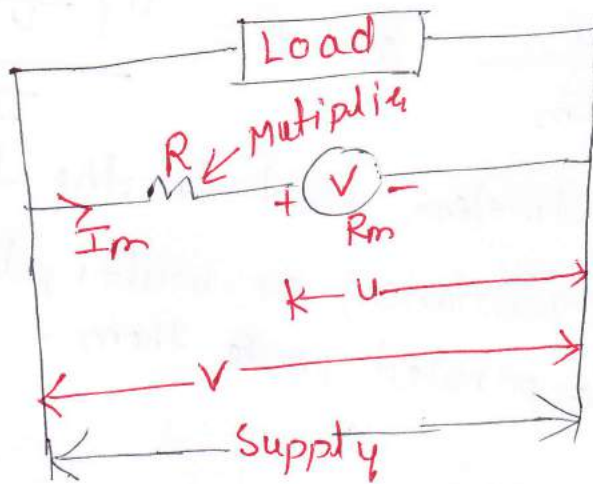


Fig → Circuit for extension of voltmeter Range.

V = Voltage to be measured

v = Voltage across the meter

I_m = full-scale deflection current of voltmeter

R = Resistance in series with the coil to extend the Range

R_m = Voltmeter Resistance

from the circuit,

$$V = I_m (R + R_m)$$

$$R + R_m = \frac{V}{I_m} \quad \text{or} \quad R = \frac{V}{I_m} - R_m$$

Required Resistance

20 ~~44~~

$$R = \frac{\text{Maximum Voltage to be measured}}{\text{full-scale deflection current of voltmeter}} - \text{Voltmeter Resistance}$$

from the circuit,

$$R = \frac{V - V_m}{I_m} \quad \text{or} \quad R = \frac{V \left(\frac{V_m}{V} - 1 \right)}{I_m}$$

→ The ammeter or voltmeters used in the laboratory etc. have shunt (low Resistance) or multiplier (high resistance) incorporated in them.

Numericals:-

Ques:- 1 The full-scale deflection current of a meter is 1mA and its internal resistance is 100 Ω . If this meter is to have full-scale deflection when 100 V is measured. What should be the value of series resistor?

Solⁿ:-

Instrument Resistance, $R_m = 100 \Omega$

full-scale deflection current, $I_m = 1 \text{ mA} = 1 \times 10^{-3} \text{ A}$

Voltage to be measured, $V = 100 \text{ V}$

value of series Resistor, $R = \frac{V}{I_m} - R_m$

$$= \frac{100}{1 \times 10^{-3}} - 100 = 99,900 \Omega$$

Q:2 The electrical lab of a polytechnic has a 0-200V range voltmeter with $20,000 \Omega/V$ sensitivity. What modifications needed to extend the range of meter to 0-2,000V? What should be the power Rating of Resistor used?

Sol: full scale deflection voltage, $V = 200V$

$$\text{Resistance of meter, } R_m = 20,000 \times 200 = 4 \times 10^6 \Omega$$

$$\text{Meter Current, } I_m = \frac{V}{R_m} = \frac{200}{4 \times 10^6} = 50 \times 10^{-6} A$$

$$\text{Voltage to be measured, } V = 2,000V$$

$$\begin{aligned} \text{Voltage across series Resistor} &= 2,000 - 200 \\ &= 1,800V \end{aligned}$$

$$\text{Resistance of the Multimeter} = \frac{1800}{50 \times 10^{-6}} = 36 \times 10^6 \Omega$$

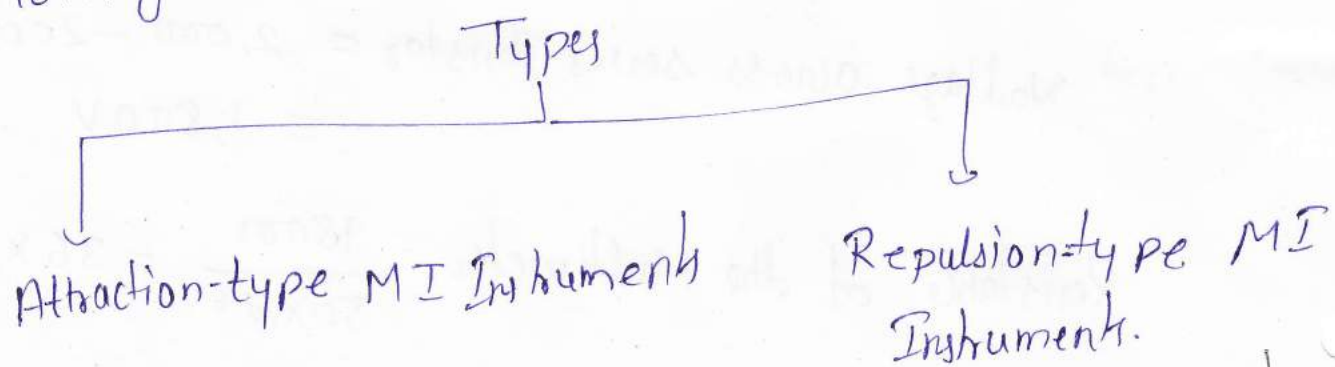
$$\begin{aligned} \text{Power Rating of the Resistor} &= I_m^2 R = \frac{(50 \times 10^{-6})^2 \times 36 \times 10^6}{36 \times 10^6} \\ &= 0.09W \end{aligned}$$

Note: Do All the Numericals from J.B Gupta Related to this topic.

⊗ Moving Iron Instruments:

(22)

- These instruments are quite cheap in cost, simple in construction & reasonably accurate at fixed power supply frequency.
- These Instruments can be used both on AC & DC.
- These are widely used in laboratories and on switching panels.
- These instruments are used either as voltmeter or ammeter only.
- Moving Iron instruments are of two types



① Attraction-type Moving Iron (MI) Instruments:

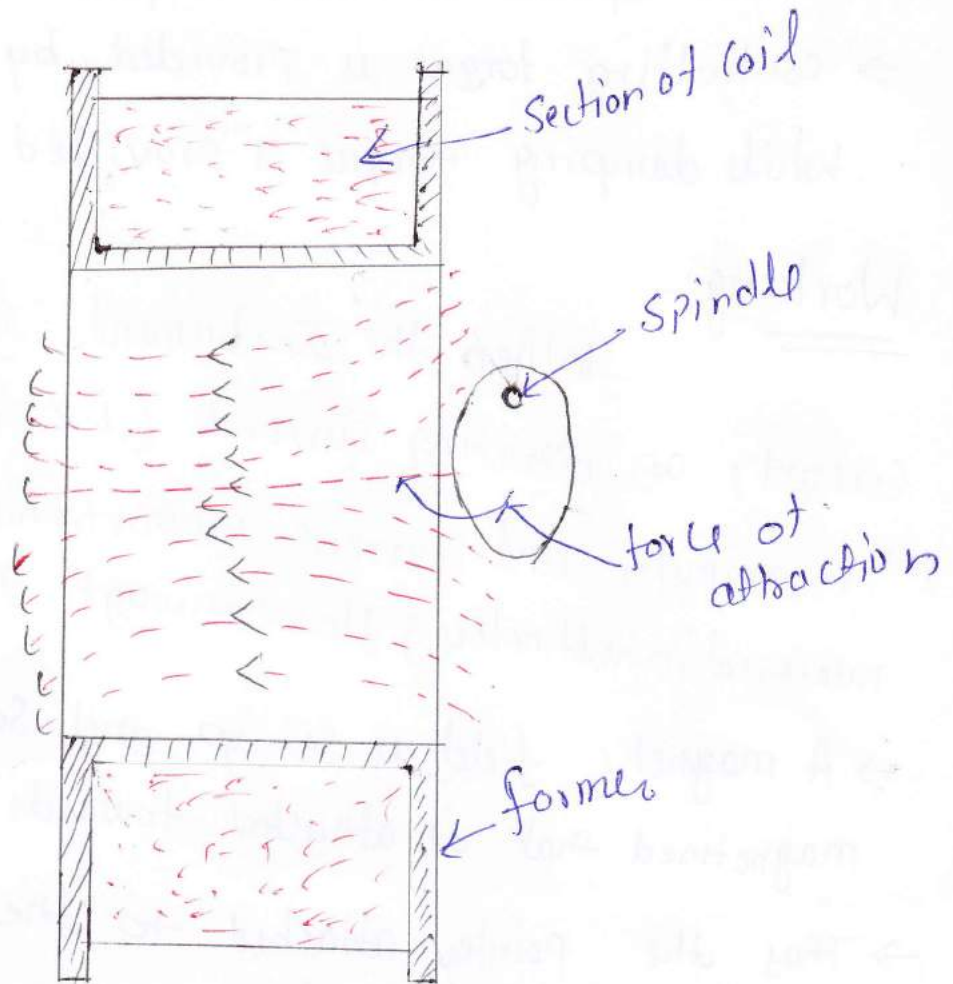
Principle:-

When a soft iron piece (or vane) is placed in the magnetic field of a current-carrying coil, it is attracted towards the centre of the coil.

- This is because the piece tries to occupy a position of minimum Reluctance. Thus a force of attraction is exerted on the soft iron piece &

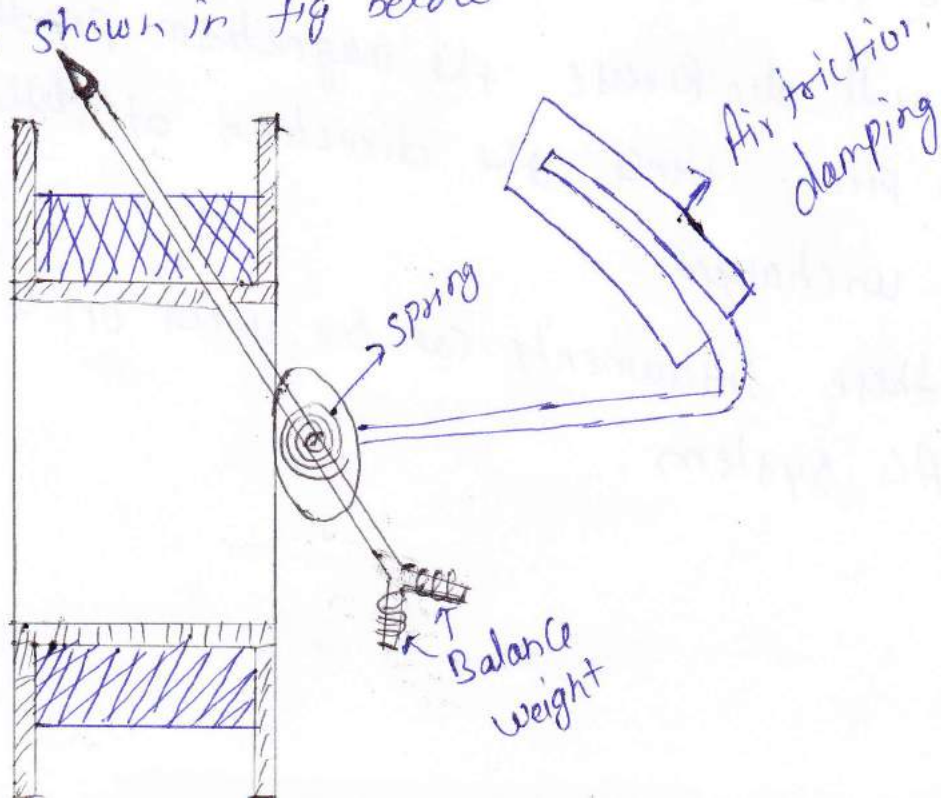
and deflection in the needle takes place. Hence the name is attraction-type Moving Iron Instruments. (23)
(I)

fig → Attraction type
Moving Iron
Instruments.



Construction:-

The sectional view of an attraction-type Moving Iron Instrument is shown in fig below.



- It consists of a stationary hollow cylindrical coil.
- An oval-shaped soft iron piece is mounted centrally to the spindle to which a pointer (needle) is attached.
- Controlling torque is provided by spring control method.
- While damping torque is provided by air friction

Working:-

- When the Instrument is connected to the circuit, an operating current (i.e. current to be measured in ammeter and current proportional to voltage to be measured in voltmeter) flows through the stationary coil.
- A magnetic field is set up and soft iron piece is magnetised that is attracted towards the centre of the coil.
 - Thus the pointer attached to the spindle is deflected over the calibrated scale.
 - If current in the coil is reversed, the direction of magnetic field produced by the coil will reverse.
 - This will also reverse the magnetism produced in the soft iron piece. Hence, the direction of deflection torque remains unchanged.
 - Thus, these instruments can be used on DC as well as on AC system.

Deflecting torque:-

(23) II

The deflecting torque T_d depends upon the force acting on the soft iron piece.

Let

H = field strength produced by the coil

m = pole strength of the soft iron piece &

$$m \propto H$$

pulling force acting on the movable iron piece

$$F \propto m \times H$$

$$F \propto H^2$$

$$(\because m \propto H)$$

Now, $H \propto I$

$$F \propto I^2$$

$$T_d \propto F$$

$$T_d \propto I^2$$

$$(T_d = \text{deflecting torque})$$

The controlling torque T_c is provided by the spiral spring

$$T_c \propto \theta$$

$$(\because \theta \rightarrow \text{angle of deflection})$$

In steady position of deflection

$$T_c = T_d$$

$\therefore \theta \propto I^2$ (I is rms value of current in AC) (24)

$\therefore$ Since, $\theta \propto I^2$, the scale of such an instrument is non-uniform, being crowded in the beginning.

Repulsion-type Moving Iron Instruments:-

Principle:-

The basic principle of a Repulsion-type M.I. Instruments is that the Repulsive forces will act b/w two similarly magnetised iron pieces when placed near to each other.

Construction:-

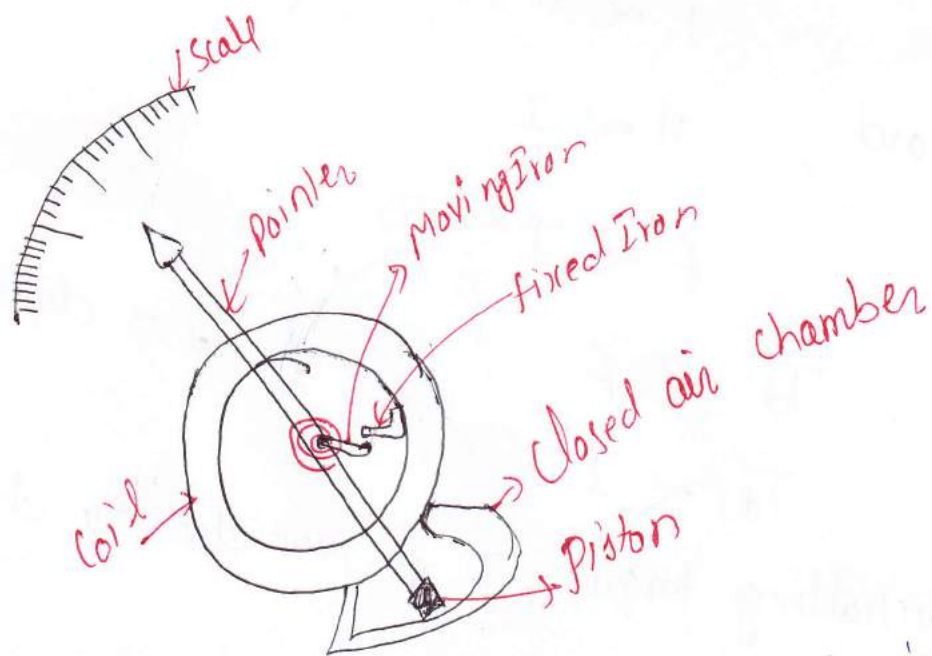


fig $\rightarrow$ Top view of Repulsion type Moving Iron Instrument

- It consists of a fixed cylindrical hollow coil that carries the operating current. (25)
- Inside the coil, there are two soft iron pieces placed parallel to each other and along the axis of the coil.
- One of the rod is fixed and the other is movable connected to the spindle.
- A pointer is attached to the spindle that gives deflection on the scale.
- The controlling torque is provided by spring control method, while damping torque is provided by air friction.

Working:- When the Instrument is connected in the circuit, the operating current flows through the coil. A magnetic field is set up along the axis of the coil.

- This field magnetises both the iron pieces and therefore both the pieces attain similar polarities.
- A force of Repulsion acts b/w the two; therefore movable piece moves away from the fixed piece. Thus the pointer attached to the spindle deflects over the calibrated scale.

- If current in the coil is Reversed, the deflection of magnetic field produced by the coil is Reversed. (26)
- Although the polarity of the magnetised soft iron pieces is Reversed but still they are magnetised similarly and Repel each other.
- Hence, the direction of deflection torque remains unchanged.
- Thus, these Instruments can be used on DC as well as on AC system.

Deflecting Torque:-

→ The deflecting torque depends upon the Repulsive force acting between the similarly magnetised iron piece. Let

H = field strength produced by the coil

m_1 = Pole strength of the fixed iron piece; ($m_1 \propto H$)

m_2 = " " " " movable " " , ($m_2 \propto H$)

Repulsive force acting on the Movable iron piece

$$f \propto m_1 m_2 \quad \text{or} \quad f \propto H^2$$

$$H \propto I$$

$$f \propto I^2$$

$$\therefore T_d \propto f$$

$$T_d \propto I^2$$

A) steady position of deflection

$$T_d \propto T_c$$

$$\theta \propto I^2$$

$\therefore \theta \propto I^2$, & therefore, the scale of such an Instrument is non-uniform, being crowded in the beginning.

Advantages:-

- 1) Cheap in cost, mechanically robust & simple in construction
- 2) Can be used on both AC & DC
- 3) They are Reasonably accurate
- 4) They possess high operating torque
- 5) They can withstand overload momentarily.

Disadvantages:-

- 1) They have non-uniform scale crowded at the beginning & therefore, it is difficult to get accurate reading at this end
- 2) Power consumption is quite high
- 3) Errors are introduced due to change in frequency in the case of AC Measurement

Errors:-

(28)

- 1) Error due to hysteresis
 - 2) " " " stray magnetic field
 - 3) Error due to temperature
- } Errors with Both AC & DC

Errors with AC only:-

- 1) Errors due change in frequency

Ranges:-

- 1) Ammeter $\rightarrow$ 0-20mA to 0-800, maximum without current transformer
- 2) Voltmeter $\rightarrow$ 0-1V to 0-800V, " " potential transformer

Applications:-

- $\rightarrow$ The Moving iron Instruments are used as ammeters & Voltmeters only.
- $\rightarrow$ The instruments can work on both AC & DC system.

⊗ Dynamometer - type Instruments:

(29)

- These are known as electrodynamic Instruments.
- These are basically Moving Coil Instruments.
- In these Instruments the operating field is produced by the fixed coil instead of permanent Magnet.

Dynamometer - type wattmeter:-

An Indicating Instrument used to Measure Power in an electric circuit is called wattmeter. It can be used on DC as well as on AC supply system.

Principle:-

- When a current-carrying Moving coil is placed in the magnetic field produced by the current-carrying fixed coil, a force is exerted on the coil & sides of the Moving coil & deflection takes place.

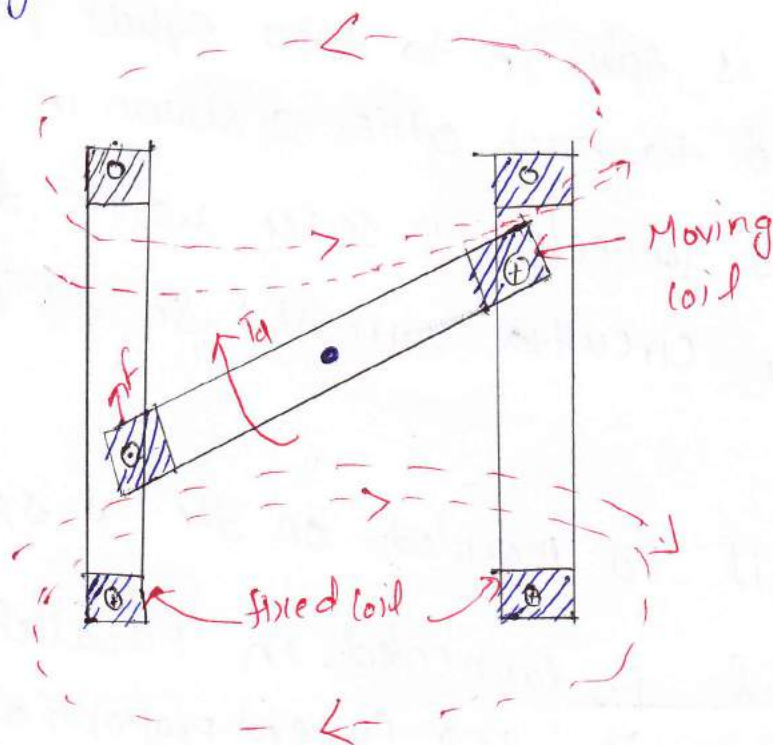


fig → working principle of wattmeter

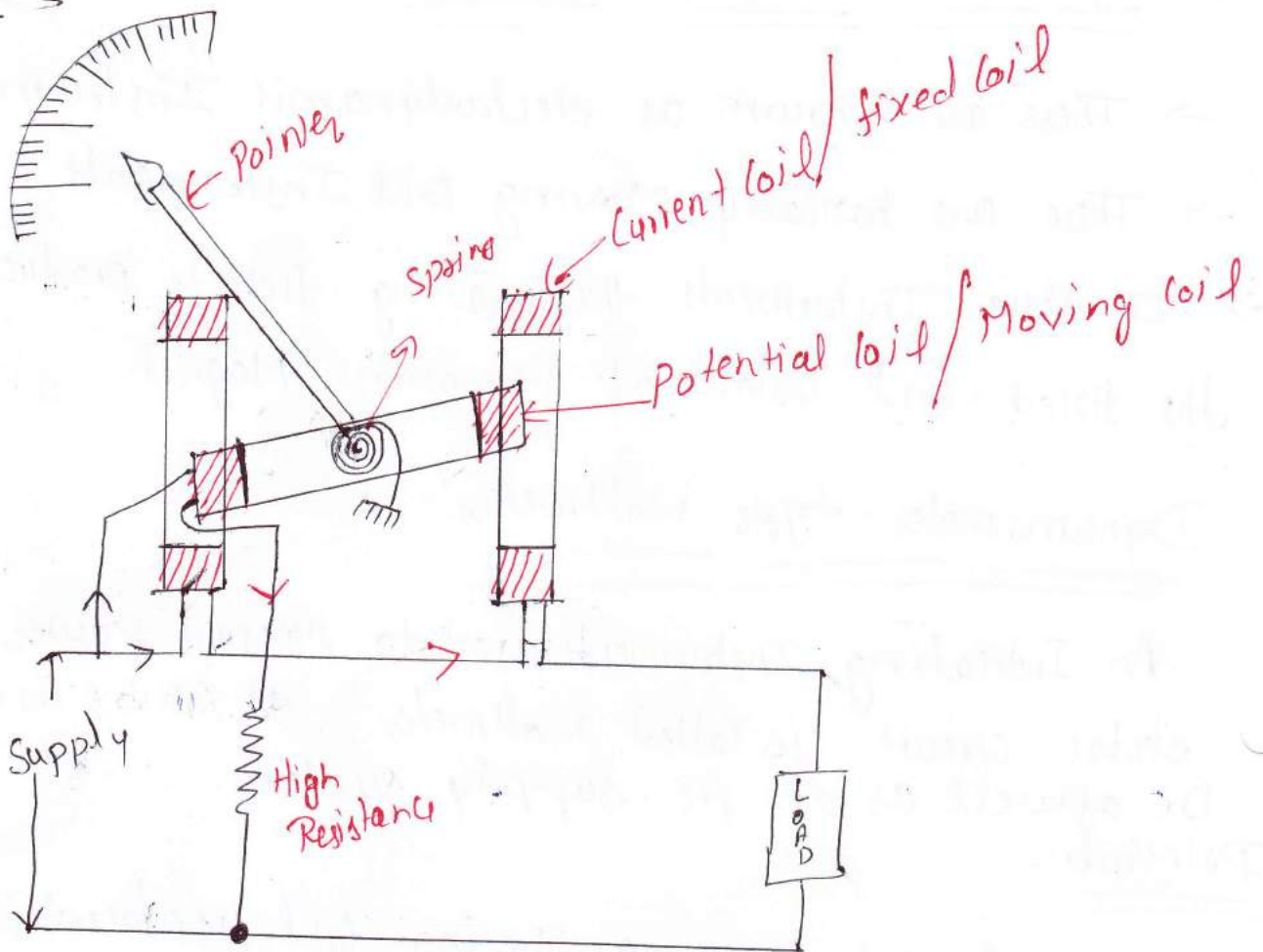


fig → circuit diagram of a Wattmeter

- The dynamo-meter type wattmeter essentially consists of two coils called fixed coil & Moving coil.
- The fixed coil is split in to two equal parts that are placed parallel to each other as shown in fig above.
- fixed coil is connected in series with the load and carries the circuit current. So, it is called Current coil
- The moving coil is mounted on the spindle.
- Moving coil is connected in parallel with the load & carries the current proportional to

the voltage applied across the load. $S, 9 + 15$ (31)
Called potential coil.

- Controlling torque is provided by the springs
- Air friction damping is used (not shown in fig.)

Working:-

- Fixed coil connected in series with load & moving coil connected in parallel with load.
- ~~→ Assume it~~
- Consider, fixed coil produces a field f_m & moving coil produces a field f_r .
- This field f_r tries to come in line with the main field f_m , which produces a deflecting torque on the moving coil.
- Thus, the pointer attached to the spindle of the moving coil deflects.
- The deflection is controlled by controlling torque produced by the springs.

Deflecting torque:

- 1) In the case of DC system
Let V = voltage across the load
 I = load current

Current through the fixed coil, $I_1 \propto I$
 " " " Moving " , $I_2 \propto V$ { Already discussed in Page 30 } (32)

So that

$$B \propto I_1 \quad (B = \text{flux density})$$

$$\therefore T_d \propto B I_2 \propto I_1 I_2 \propto I V \propto \text{Power}$$

$$\therefore T_d \propto \text{Power}$$

Hence, deflecting torque is proportional to power

(2) In the case of AC system

$$T_d \propto V I \cos \phi$$

$$T_c \propto Q$$

$$\therefore T_c = T_d$$

$$Q \propto V I \cos \phi \propto \text{Power}$$

Advantages:-

- 1) It can be used on DC as well as AC circuit
- 2) It has almost uniform scale
- 3) High degree of accuracy can be obtained

Disadvantages:-

- 1) At low power factors, the Inductance of the potential coil cause serious errors.
- 2) The reading of the Instrument may be affected by ~~Star~~ Stray fields acting on the Moving coil.

Errors:-

- 1) Error due to potential coil Inductance
- 2) Effect due to power loss in potential coil or current coil
- 3) Error due to eddy currents
- 4) Error due to stray magnetic field.

Range:-

- 1) Current circuit 0-0.25 A to 0-100 A, without using current transformer
- 2) Potential circuit 0-5 A to 0-750 V, without using potential transformers.

⊗ Induction-type Instruments:-

(34)

→ Induction-type Instruments are used only for AC measurements.

→ These instruments can be used either as ammeter, voltmeter or wattmeter.

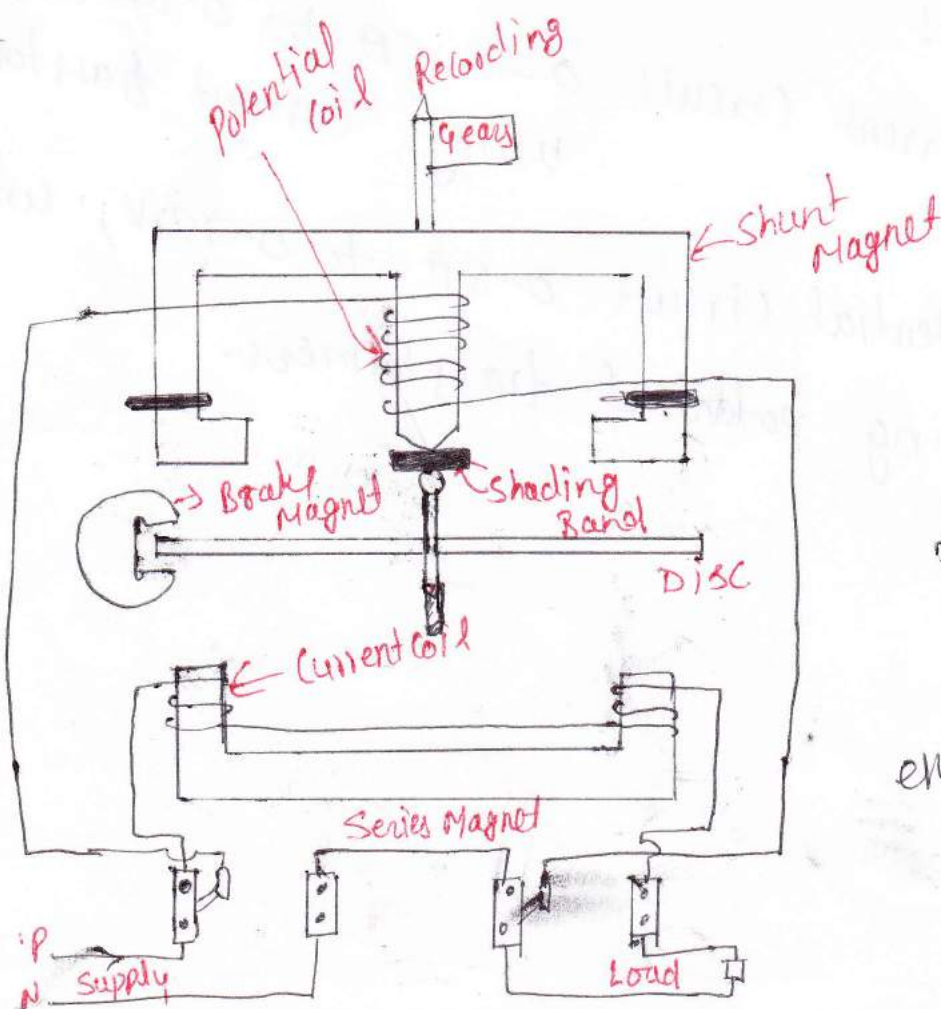
Induction-type Single-phase Energy Meter

Although Induction-type Instruments can be used as ammeter, voltmeter or wattmeter, the Induction-type energy meters are more popular.

Principle:-

The basic principle of Induction-type energy meter is Electromagnetic Induction.

Construction:-



Arg's connection
of CC &
PC in an
energy meter.

Construction:-

(35)

An Induction-type single-phase energy meter has the following main parts of the operating mechanism

1) Driving System:- The driving system of the meter consist of two electromagnets.

a) Series Magnet:- It consist of number of V-shaped laminations of silicon steel staggered together to form a core.

→ A coil of thick wire having a few turns is wound on ~~the~~ both the legs of V-shaped magnet.

→ This coil is connected in series with the load.

b) Shunt Magnet:-

→ It consists of a number of M-shaped lamination of silicon steel assembled together to form a core.

→ A coil of thin wire having large number of turns is wound on the central limb of the magnet.

→ This coil is connected across the load.

→ Copper Shading Band (Short circuiting copper ring) is provided on the central limb of the shunt

Magnet. The phase difference of 90° is obtained by adjusting the position of shading Band.

2) Moving System:-

(36)

It consist of a light Al disc mounted on a vertical spindle.

3) Braking System:-

→ A permanent Magnet positioned near the edge of the Al disc. The position of Braking magnet is adjustable.

4) Recording Mechanism:-

→ The function of Recording or Registering Mechanism is to record continuously a number on the dial that is proportional to the Revolutions made by the moving system.

Working:-

→ When the energy meter is connected in the circuit the current coil carries the load current & Pressure coil carries the current proportional to the Supply Voltage.

→ The magnetic field produced by the series magnet is in phase with the line current & the magnetic field produced by the shunt magnet is in quadrature with the applied voltage.

→ Thus, a phase difference exists b/w the fluxes produced by the two coils. This sets up a rotating field that interacts with the disc and produces a driving torque and thus the disc starts rotating. (37)

→ The number of Revolutions made by the disc depends upon the energy passing through the meters.

→ The spindle is geared to the Recording Mechanism so that electrical energy consumed in the circuit is directly ~~registered~~ registered in kWh.

→ The Speed of the disc is adjusted by adjusting the position of the Braking magnet.

→ for example → If the energy meters registers less energy than the energy actually consumed in the circuit, then the speed of the disc has to be increased that is obtained by shifting the braking magnet near to the Centre of the disc & vice-versa.

Errors:-

Deflecting torque:-

$$T_d \propto \text{Power}$$

$$T_B \propto N \quad \because \quad T_B \rightarrow \text{Braking Torque}$$

$$N \rightarrow \text{Speed of the disc}$$

for steady speed ,

$$T_d = T_B \quad \text{--- eqn (2)} \quad \left(T_d = \text{driving Torque} \right)$$

from eqn (1) & (2),

$$\textcircled{1} \quad N \propto P \quad , \quad (P \text{ is power})$$

Multiplying both sides by time t , we get

$$Nt \propto Pt$$

Since the product Nt represents the total number of Revolution of the disc in time t & the product Pt represents the energy passing through the meter in time t .

→ Thus the number of Revolutions of the disc are directly proportional to electrical energy passing through the meter

Errors:-

- 1) phase & speed error
- 2) frictional error
- 3) Creeping error
- 4) Temperature error
- 5) frequency error